Vikram Anantha

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Education

University of Michigan — Aug. 2024 – May 2028 (expected)
B.S. Computer Science & Electrical Engineering, 1931E Scholar — *3.96/4.00 GPA*

- Relevant coursework: Dynamics, Logic Design, Machine Learning, Computer Vision, Data Structures & Algorithms, Digital Circuits, Computer Organization, Linear Algebra, Discrete Math, Diff Eqs
- Clubs: Photo Editor of the Michiganensian Yearbook

Industry Experience

Capoom — Jan. 2025 – Present
Software Eng. Intern — *Ann Arbor, MI*

- Architected an automated pipeline for 3D Gaussian Splat model creation of UofM's Mcity
- Implemented hierarchical methods to visualize Gaussian Splat map

Stellantis — May 2025 – Aug. 2025
ADAS Functional Integration Eng. Intern — *Auburn Hills, MI*

- Deployed a CAPL script to diagnose CAN-bus anomalies, identifying a 10% vehicle-component fault rate
- Designed and validated MATLAB Simulink model to control CAN-bus signals of vehicle to quantify the self-driving feel

Constant Therapy Health — Jun. 2024 – Aug. 2024
Software Eng. Intern — *Newton, MA*

- Developed Automatic Speech Recognition model for stroke aphasia patients & web-based hierarchical task visualizer
- Reduced false negative rate by 4x compared to Google's Speech Recognition Engine

HELM Learning — Apr. 2020 – May 2024
Co-Founder, Teacher, Developer — *Lexington, MA*

- Created free online peer-to-peer worldwide learning platform with over 4.2K students, 90+ classes
- Created services with Python, JS, SQL, Recommendation ML Algorithm, hosted platform on AWS

Research Experience

University of Michigan, ROAHM Robotics Lab — Aug. 2024 – Present

- Creating a real-time 3D object segmented model of a changing env. in Gaussian Splat for robotic arm movement
- Developing new Transformer-based Vision Language Action models to create faster generalist robot policies

Columbia University, Center for Smart Street Scapes — May 2023 – Feb 2024

- Worked on NSF Project, developed app to warn pedestrians at road intersections of oncoming vehicles
- Reduced latency to ¡2s for human reaction time to prevent accidents using React Native and MQTT messaging

Massachusetts Institute of Technology, Auto ID Lab — Jun. 2023 – Sep. 2023

- Worked on 2FA system to prevent hacking for auton. vehicles on roads with YOLOv8 Vision model and Python

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, CAPL, ReactJS, Assembly, Bash, Verilog
Frameworks & Tools: Robot Operating System (ROS & ROS2), Amazon Web Services (AWS), Vector CANalyzer, MATLAB/Simulink, OpenCV, TensorFlow/PyTorch, Git/GitHub, CUDA, Docker, Gaussian Splatting, LLMs, YOLOv8 Object Detection, SAM-2 Segmentation, SLAM

Selected Honors & Awards

Immerse The Bay Hackathon, Stanford University, *1st Place Winner* — Nov 2025
1931e Merit Scholarship, University of Michigan, *1 of 6 recipients* — May 2025
American Invitational Mathematics Exam (AIME), *2 Time Qualifier* — Mar. 2021 & Feb. 2022
USA Coding Olympiad (USACO), *Silver Division* — Dec. 2019

Publications

Digital Twin for Pedestrian Safety Warning at a Single Urban Traffic Intersection — 2024

- V. Anantha, et al., 2024 IEEE Intelligent Vehicles Symposium (IV), pp. 2640-2645, doi: 10.1109/IV55156.2024.10588544

Peer to Peer Learning Platform Optimized With Machine Learning — 2022

- V. Anantha, arXiv:2209.03489, 2022